Amrithya Balaji

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Education

MSc Machine Learning and Data Mining, University of Jean Monnet — Aug 2023 – Sep 2025

MSc Computer science ERASMUS+ semester, University of Freiburg — Oct 2024 – Mar 2025

B.E Computer Science and Engineering, SVCE, Chennai, India — Aug 2017 – May 2021

Experience

AI Researcher, Intern, IBDM, Marseille — Apr 2025 – Aug 2025

- Designing interpretable large language models (LLMs) to classify cellular states from bulk and single-cell RNA-seq data. Integrating explainability frameworks (e.g., SHAP, attention maps) to identify key genes influencing cell differentiation. improved classification accuracy and biological interpretability by >5% vs baseline models.
- **Reference**: Ronan Sicre, Université Paul Sabatier Toulouse III, Junior Professor Chair, **ronan.sicre@lis-lab.fr** and Bianca Habermann, Group Leader Computational Biology, IBDM, **bianca.HABERMANN@univ-amu.fr**

Student Researcher, Part-Time, Bioinformatics Group, Albert-Ludwigs-University – Freiburg, Germany — Nov 2024 – Mar 2025

- Built a BERT and GRU based deep learning pipeline predicting neoantigen immunogenicity on experimentally validated datasets. Achieved 80% classification accuracy.

ML Researcher, Intern, Laboratoire Hubert Curien – Saint Etienne — Apr 2024 – Jul 2024

- Enhanced graph learning models by incorporating screening-based sparsity methods, reducing runtime by 10–20%. Contributed to research on interpretable structured learning under varying optimization constraints.
- **Reference:** Benjamin Girault, Researcher, INRIA, **benjamin.girault@inria.fr**

Programmer Analyst, Cognizant Technology Solutions – Chennai, India — Jul 2021 – Aug 2023

- Engineered and deployed full-stack web applications using Angular, JSP, and SQL Server for enterprise clients. Collaborated across 4 agile teams as DevOps Developer managing databases (Postgres, MongoDB) and build pipelines.

Projects: **github.com/amrithya**

Post-hoc Explainable AI (XAI) for Protein Subcellular Localization (in progress)

Fine-tuned the pretrained 'cytoself' model on human protein images. Currently working on applying post-hoc XAI techniques including latent-space visualization and feature attribution to interpret model predictions.

Neoantigen Immunogenicity Prediction

Improved classification accuracy by training BERT based architecture on 20k+ sequence data samples over GRU-based classifier.

Structured Generation from Call Transcripts using FastAPI and LLaMA

Developed a web application to generate call transcripts and extract structured information from them. Built the backend API using FastAPI and used LLaMA / OpenAI API with prompt engineering for transcript generation and information extraction.

Designed an interactive frontend using HTML, CSS, and JavaScript for user inputs and results display

Transfer Learning using ResNet

Fine-tuned a pretrained ResNet model on the MNIST dataset for digit classification.

Library Management System with Book Recommender

Developed an end-to-end application with a collaborative filtering-based book recommendation system. Built the backend using Flask and designed interactive frontend interfaces using JavaScript.

Skills

Programming Languages: C, C++, Java, Python, Angular
Data Science: Data Collection, Data Processing, Data Analysis, Data Visualization, Machine Learning, Artificial Intelligence, Natural language processing (NLP) – LLMs, Computer vision, RAG
Deep Learning frameworks: PyTorch, Tensorflow, Keras
Frameworks & Libraries: OpenCV, Sci-kit learn, NumPy, Pandas, SciPy, Matplotlib, Flask
Tools: Git, MSOffice, Maven, Udeploy, Apache Subversion, Docker, HPC, FastAPI
Natural Languages: **English:** Fluent, **French:** Beginner, **German:** Beginner, **Tamil:** Native